

30 — GENETIC ALGORITHM TO MAXIMISE A FUNCTION

Aim

To implement a simple Genetic Algorithm to maximise the function $f(x) = x^2$ for integer values of x from 0 to 31 using 5-bit binary chromosome encoding.

Algorithm + Procedure

1. Represent each candidate solution as a 5-bit binary chromosome.
2. Generate an initial population randomly.
3. Decode each chromosome into its corresponding integer value.
4. Calculate the fitness using:

$$f(x) = x^2$$

5. Calculate the selection probability of each chromosome based on its fitness.
6. Select parent chromosomes using roulette-wheel selection.
7. Perform single-point crossover to generate new offspring.
8. Apply mutation to randomly change some bits.
9. Replace the old population with the new population.
10. Repeat the selection, crossover, and mutation processes for several generations.
11. Select the chromosome with the highest fitness.
12. Decode it to obtain the best value of x .
13. Display the best x and its fitness value.

Code

```
import random

else:

    probabilities = [
        f / total_fitness
        for f in fitness_values
    ]

    new_population = []

    # Selection and crossover

    while len(new_population) < population_size:
```

```

parent1, parent2 = random.choices(
    population,
    weights=probabilities,
    k=2
)
# Single-point crossover
point = random.randint(1, 4)
child1 = (
    parent1[:point] +
    parent2[point:]
)
child2 = (
    parent2[:point] +
    parent1[point:]
)
new_population.append(child1)
if len(new_population) < population_size:
    new_population.append(child2)
# Mutation
population = []
for chromosome in new_population:
    mutated = ""
    for bit in chromosome:
        if random.random() < mutation_rate:
            bit = str(1 - int(bit))
        mutated += bit
    population.append(mutated)
# Find best chromosome
best = max(

```

```
        population,
        key=fitness
    )
    return decode(best), fitness(best)

# Run Genetic Algorithm
random.seed(1)
x, fx = genetic_algorithm()
print("Best x =", x)
print("f(x) =", fx)
```

Output

```
Best x = 31
f(x) = 961
```

Result

The Genetic Algorithm was successfully implemented using **5-bit chromosome encoding, roulette-wheel selection, single-point crossover, and mutation**. The algorithm converged to the maximum value:

$x = 31$ and $f(x) = 961$.